MGT 205 Project 3

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Run the three code blocks below to get the heart data set.

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.1.4     ✔ readr     2.1.4
## ✔ forcats   1.0.0     ✔ stringr   1.5.1
## ✔ ggplot2   3.4.4     ✔ tibble    3.2.1
## ✔ lubridate 1.9.3     ✔ tidyr     1.3.0
## ✔ purrr     1.0.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
```

```
heart <- read_csv('heart.csv')
```

```
## Rows: 303 Columns: 14
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## dbl (14): age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpea...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(heart)
```

```
## # A tibble: 6 × 14
##     age   sex    cp trestbps  chol   fbs restecg thalach exang oldpeak slope
##   <dbl> <dbl> <dbl>    <dbl> <dbl> <dbl>   <dbl>   <dbl> <dbl>   <dbl> <dbl>
## 1    63     1     3      145   233     1       0     150     0     2.3     0
## 2    37     1     2      130   250     0       1     187     0     3.5     0
## 3    41     0     1      130   204     0       0     172     0     1.4     2
## 4    56     1     1      120   236     0       1     178     0     0.8     2
## 5    57     0     0      120   354     0       1     163     1     0.6     2
## 6    57     1     0      140   192     0       1     148     0     0.4     1
## # ℹ 3 more variables: ca <dbl>, thal <dbl>, target <dbl>
```

Data Description and Usage

- **age:** The person's age in years

- **sex:** The person's sex (1 = male, 0 = female)
- **cp:** The chest pain experienced (Value 1: typical angina, Value 2: atypical angina, Value 3: non-anginal pain, Value 4: asymptomatic)
- **trestbps:** The person's resting blood pressure (mm Hg on admission to the hospital)
- **chol:** The person's cholesterol measurement in mg/dl
- **fbs:** The person's fasting blood sugar (> 120 mg/dl, 1 = true; 0 = false)
- **restecg:** Resting electrocardiographic measurement (0 = normal, 1 = having ST-T wave abnormality, 2 = showing probable or definite left ventricular hypertrophy by Estes' criteria)
- **thalach:** The person's maximum heart rate achieved
- **exang:** Exercise induced angina (1 = yes; 0 = no)
- **oldpeak:** ST depression induced by exercise relative to rest ('ST' relates to positions on the ECG plot. See more here)
- **slope:** the slope of the peak exercise ST segment (Value 1: upsloping, Value 2: flat, Value 3: downsloping)
- **ca:** The number of major vessels (0-3)
- **thal:** A blood disorder called thalassemia (3 = normal; 6 = fixed defect; 7 = reversable defect)
- **target:** Heart disease (0 = no, 1 = yes)

Input code for and answer all prompts below.

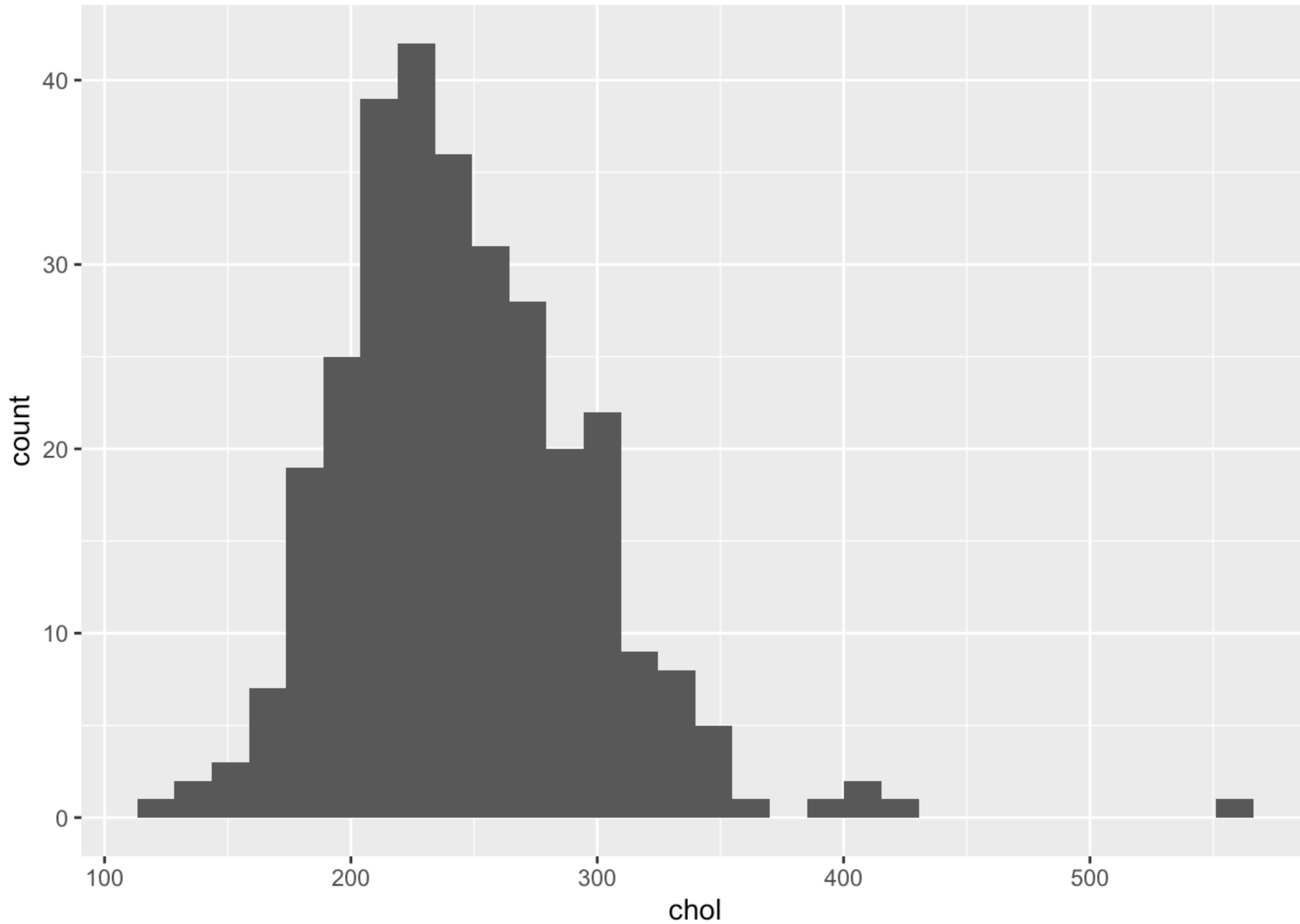
Initial Investigation

Question 1

1a. Using the chol variable, plot the distribution as a histogram.

```
heart %>%
  ggplot(aes(x = chol)) +
  geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



1b. Describe the distribution's shape (e.g. left or right skew, normal, etc).

```
#The distribution's shape is right skewed
```

Question 2

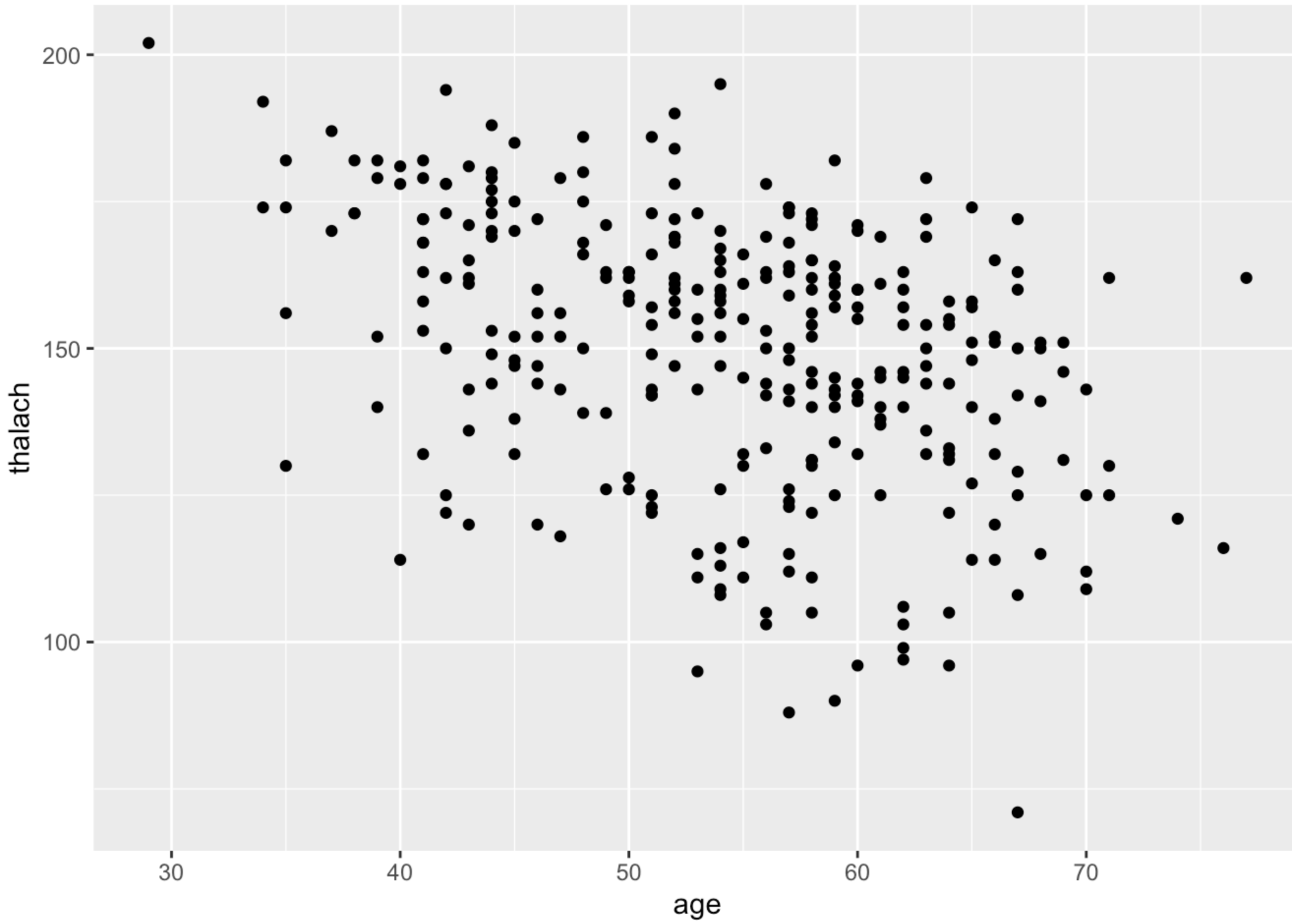
2a. Using the age and thalach variables, find their correlation coefficient.

```
cor <- cor(heart[,1],heart[,8],use='complete',method='pearson')
cor
```

```
##        thalach
## age -0.3985219
```

2b. Show this relationship in a plot.

```
heart %>%
 ggplot(aes(x=age,y=thalach)) +
 geom_point()
```



2c. Does this correlation make sense for this data? Why/why not?

```
#This correlation makes sense. Because as age increases, human's physical functions and
health reduce, therefore have lower heart rates.
```

Hypothesis Testing

Question 3

3a. Perform a t test for difference in means between genders (sex) on chol.

```
t.test(chol ~ sex, data = heart)
```

```
##
##  Welch Two Sample t-test
##
## data:  chol by sex
## t = 3.0244, df = 134.39, p-value = 0.002985
## alternative hypothesis: true difference in means between group 0 and group 1 is not e
qual to 0
## 95 percent confidence interval:
##   7.617474 36.406982
## sample estimates:
## mean in group 0 mean in group 1
##        261.3021        239.2899
```

3b. Find and show the mean cholesterol of men and women.

```
heart %>%
 group_by(sex) %>%
 summarise(mean_cholesterol = mean(chol))
```

```
## # A tibble: 2 × 2
##     sex mean_cholesterol
##   <dbl>            <dbl>
## 1     0             261.
## 2     1             239.
```

3c. Interpret the results above using a significance threshold of 0.05. Are the results statistically significant?

```
#Because the p-value = 0.002985 < significance threshold 0.05, the results are statistic
ally significant.
```

Question 4

4a. Perform a t test (hypothesis test) to determine if the blood pressure among those with heart disease is higher or not.

```
t.test(trestbps ~ target, data = heart, alternative = 'less')
```

```
##
##  Welch Two Sample t-test
##
## data:  trestbps by target
## t = 2.5083, df = 272.56, p-value = 0.9936
## alternative hypothesis: true difference in means between group 0 and group 1 is less
than 0
## 95 percent confidence interval:
##      -Inf 8.448315
## sample estimates:
## mean in group 0 mean in group 1
##        134.3986        129.3030
```

4b. Interpret these results. Statistically, is blood pressure higher among those with heart disease? How can you tell?

```
#The results are not statistically significant. We fail to reject the null hypothesis. T
here is insufficient evidence to conclude that resting blood pressure is higher among in
dividuals with heart disease.
```

Modelling

Question 5

5a. Separate heart data into a training and testing data set.

```
sample_indexes <- sample(1:nrow(heart), size = nrow(heart) * 3/4)
heart_train <- heart[sample_indexes, ]
heart_test <- heart[-sample_indexes, ]
```

5b. Construct a LINEAR regression model to predict for blood pressure.

```
fit <- lm(trestbps ~ ., data = heart_train)
summary(fit)
```

```
##
## Call:
## lm(formula = trestbps ~ ., data = heart_train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -34.531 -11.214  -2.583  10.220  52.703
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 82.27851   14.69977   5.597 6.66e-08 ***
## age          0.48965    0.14343   3.414 0.000767 ***
## sex         -2.98269    2.68222  -1.112 0.267383
## cp           3.09584    1.26592   2.446 0.015276 *
## chol         0.03750    0.02420   1.549 0.122797
## fbs          5.31727    3.01292   1.765 0.079026 .
## restecg     -1.60514    2.22819  -0.720 0.472082
## thalach      0.10656    0.05999   1.776 0.077120 .
## exang        0.18955    2.73809   0.069 0.944875
## oldpeak      1.06635    1.28801   0.828 0.408649
## slope       -1.47851    2.34178  -0.631 0.528479
## ca           0.57162    1.22524   0.467 0.641308
## thal         0.57033    1.91458   0.298 0.766080
## target      -6.66474    3.16091  -2.108 0.036158 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 16.26 on 213 degrees of freedom
## Multiple R-squared:  0.1971, Adjusted R-squared:  0.1481
## F-statistic: 4.023 on 13 and 213 DF,  p-value: 6.688e-06
```

5c. What are the statistically significant variables in this model? What is the value of the Multiple R-squared statistic, and what does this value mean? Overall, is this a good prediction model?

```
#1. Statistically significant variables: age (because p_value 0.000134 < 0.05) fbs (beca
use p_value 0.015681 < 0.05) oldpeak (because p_value 0.044594 < 0.05)
#2. Multiple R-squared: 0.1531 R^2=1-SSE/SST (SSE:error sum of squares, SST:total sum of
squares)
  #It refers to the goodness of fit, which is how well the regression line fits the obse
rvations. Its value is between 0 and 1, the closer to 1, the better the regression fitti
ng effect. Because Multiple R-squared is 0.1531, the regression fits not well and is not
a good prediction model.
```

Question 6

6a. Construct a LOGISTIC regression model to predict for heart disease indicator flag. Calculate the accuracy using the testing data.

```
fit_class <- glm(target ~ ., data = heart_train, family = binomial(link = 'logit'))

prediction_class <- ifelse(predict(fit_class, newdata = heart_test, type = 'response') <
0.5, 0, 1)

sum(heart_test$target == prediction_class)/length(prediction_class)
```

```
## [1] 0.8157895
```

6b. What percent of the time is this model accurate?

```
#81.57895% of the time this model is accurate
```